# Linux Command

| Category | Linux Command |
|---|---|
| System Information & Hardware | uname -a |
| System Information & Hardware | hostname |
| System Information & Hardware | uptime |
| System Information & Hardware | top / htop |
| System Information & Hardware | df -h |
| System Information & Hardware | free -h |
| File & Directory Management | pwd |
| File & Directory Management | ls -la |
| File & Directory Management | cd [directory] |
| File & Directory Management | mkdir [name] |
| File & Directory Management | cp [src] [dst] |
| File & Directory Management | mv [src] [dst] |
| File & Directory Management | rm [file] |
| File & Directory Management | cat [file] |
| File & Directory Management | less [file] |
| File & Directory Management | head / tail |
| User, Group & Permission Management | whoami |
| User, Group & Permission Management | sudo [cmd] |
| User, Group & Permission Management | useradd -m [user] |
| User, Group & Permission Management | passwd [user] |
| User, Group & Permission Management | groupadd [group] |
| User, Group & Permission Management | usermod -aG [grp] [usr] |
| User, Group & Permission Management | chmod [perms] [file] |
| User, Group & Permission Management | chown [usr:grp] [file] |
| Networking Commands & Diagnostics | ip a / ifconfig |

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| Networking Commands & Diagnostics | ip route |
| Networking Commands & Diagnostics | ping [target] |
| Networking Commands & Diagnostics | netstat -tulnp / ss -tulnp |
| Networking Commands & Diagnostics | traceroute [target] |
| Networking Commands & Diagnostics | dig / nslookup |
| Search, Filtering & Text Processing | find [path] [opts] |
| Search, Filtering & Text Processing | grep "[text]" [file] |
| Search, Filtering & Text Processing | sort |
| Search, Filtering & Text Processing | uniq |
| Search, Filtering & Text Processing | strings |

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| What It Does |
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| Displays complete system and kernel information. |
| Shows or sets the system's host name. |
| Shows how long the system has been running, user count, and load averages. |
| Displays active system processes, CPU usage, and memory consumption in real-time. |
| Shows available and used disk space on all mounted filesystems in human-readable format. |
| Displays total, used, and free system RAM and swap space. |
| Prints the absolute path of the current working directory. |
| Lists all files (including hidden files) with detailed permissions, owners, and sizes. |
| Changes the current working directory to the specified path. |
| Creates a new directory. |
| Copies files or directories from source to destination. |
| Moves or renames files and directories. |
| Deletes files or directories. |
| Displays the entire content of a file directly on the terminal screen. |
| Opens a file interactively with scrolling capability. |
| Displays the first or last 10 lines of a file (default). |
| Displays the username of the currently logged-in user. |
| Executes a command with administrative (root) privileges. |
| Creates a new user account along with a home directory. |
| Sets or changes the password for a user account. |
| Creates a new user group. |
| Adds an existing user to a specific secondary group. |
| Changes read, write, and execute permissions of a file or directory. |
| Changes the user owner and group association of a file. |
| Displays network interfaces, assigned IP addresses, and MAC configurations. |

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| Displays the system routing table and default gateway. |
| Sends ICMP echo requests to test network reachability to another host. |
| Lists all active listening ports and the associated programs running them. |
| Traces the network path packets take to reach a destination host. |
| Queries DNS servers to resolve domain names to IP addresses. |
| Searches for files and directories based on name, size, or permissions. |
| Searches for specific text patterns or keywords inside files or streams. |
| Sorts lines of text files in alphabetical or numerical order. |
| Filters or reports repeated lines in a text file. |
| Extracts human-readable ASCII text strings from binary files. |

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| When to Use It |
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| To check the Linux kernel version and architecture during system enumeration. |
| To verify which machine or virtual environment you are currently working on. |
| To check system stability and how long a server has been active. |
| To monitor resource utilization and identify misbehaving or suspicious processes. |
| To check if a virtual machine or partition is running out of disk storage. |
| To check available system memory during heavy tool executions. |
| To verify your exact location within the Linux file system hierarchy. |
| To inspect directory contents and spot hidden configuration files or directories. |
| To navigate between folders within the terminal. |
| To organize your lab files, writeups, and scripts into folders. |
| To duplicate configuration files or backup important scripts before editing. |
| To reorganize files or rename script extensions. |
| To remove temporary or unnecessary files from your workspace. |
| To quickly read text logs, short scripts, or configuration files. |
| To read long log files or text documents page by page. |
| To quickly check the beginning or recent updates of a log file (e.g., tail -f /var/log/auth.log). |
| To verify whether you are operating as a standard user or root. |
| To perform system tasks that require elevated permissions. |
| To add new users to the Linux system. |
| To secure user accounts with a strong password. |
| To organize users into specific functional groups. |
| To grant group-level access permissions to a user. |
| To secure script execution or restrict unauthorized file modifications. |
| To assign proper file ownership for system services or applications. |
| To check your machine's IP address and network interface status. |

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| To troubleshoot local network routing and gateway connectivity. |
| To verify if a target machine or external website is online and reachable. |
| To check which services are running on specific ports and detect unauthorized listeners. |
| To analyze network latency and routing bottlenecks across routers. |
| To investigate DNS records, domain resolutions, or potential DNS hijacking. |
| To locate specific files, SUID binaries, or hidden directories across the system. |
| To extract specific error messages, usernames, or indicators of compromise from logs. |
| To organize data lists during log analysis or text processing. |
| To identify unique entries or count duplicates in log files. |
| To analyze compiled binaries or malware files for hidden messages or URLs. |

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